SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Ashwanth SP/192126001 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ashwanth SP

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Types of Intelligent Agents

1. Simple reflex Agent

- Works using condition-action rule.
- Responds only to the current input.
- Example: If motion is detected, switch on the lights.
- Advantage: fast and simple.
- Disadvantage: Required memory

2. Model-Based agent:

- Maintain an internal model of the environment.
- Eg: Remembers room occupancy and adjust temperature.
- Advantage: Handles changing environments.
- Disadvantage: Requires memory.

3. Goal-Based Agent:

- Takes actions to achieve specific goals.
- Eg: Maintain a comfortable room temperature.
- Advantage: Plans actions to reach goals
- Disadvantage: Requires more computation.

Best approach:

A hybrid approach combining model-based and utility based agents is most effective. It learns user behavior and balances comfort, energy efficiency and security. A simple reflex component can respond quickly to emergencies..

1. Uniform Cost Search (UCS)

UCS explores the path with the lowest total cost first.

Steps:

1. Start from the initial position.
2. Explore neighboring paths.
3. Select the path with the lowest cost.
4. Repeat until the exit is found.

Advantage: Finds the best-cost path when step costs are non-negative.

Disadvantage: Requires more time and memory.

Complexity: Time and space can be exponential in the worst case.

Iterative deepening search (IDS)

It systematically combines depth-limited search with depth limit each time.

Steps:

1. Start with depth limit 0.
2. Increase the limit gradually.
3. Explore path at each level
4. Stop when the goal is found.

Advantages: Uses less memory and finds the shallowest solution when step costs are equal.

Disadvantage: Repeats searches and does not guarantee the best cost path for different movement costs.

Complexity:

- Time: $O(b^d)$
- Space: $O(bd)$

Comparison:

Feature	UCS	IDS
Search	Lowest cost first	Increasing depth
Memory	High	Low
Goal, cost Path Optimality	Least-cost path	Shallowest path for equal costs.

Intelligent agents and best approach:

* Simple reflex: Reacts to nearby obstacles.
* Model-based: Remembers visited locations and builds a map.
* Goal-based: plans a path to the exit.
* Utility-based: chooses paths based on cost, energy, and safety

conclusion:

* model-based, goal-based agent using IDS is suitable for equal-cost paths and limited memory for different movement costs, UCS is better because it finds the least-cost path.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand